This material was developed in the context of the Nordplus project NPHE-2025/10158, “Co-designing a Computational Thinking and Mathematics Curriculum Framework for Nordic-Baltic Preservice Teacher Education

Checklist for Designing Computational Thinking–Oriented Mathematics Activities for Preservice Teacher Education

Activity title:		**Grade level:**	
Mathematical topic:		**Programming / CT environment:**	
Designer		**Date:**	
Intended classroom use:		**Estimated duration:**	

Overall information

Begin by specifying the mathematical learning objectives and identifying the mathematical and technical prior knowledge required from PSTs. Then consider the checklist domains collectively when designing or reviewing the activity.

The teacher educator explicitly models mathematical and computational thinking by explaining the reasoning behind decisions and strategies. The following Checklist could be used in drafting activities.

Requirements	Checklist item	Yes	Partly	No	Evidence / comments
1. Mathematical focus and overall activity structure	The mathematical learning objectives are clearly specified.	☐	☐	☐	
	The mathematical and technical prior knowledge required from PSTs is identified.				
	The activity has a clear and purposeful connection to school mathematics, with the mathematical idea being central rather than serving merely as a superficial context.				
	The activity promotes stepwise and systematic problem solving.				
	The activity involves iteration or debugging-like processes.	☐	☐	☐	
	PSTs are expected to plan a solution strategy before or during implementation.	☐	☐	☐	
	The activity allows PSTs to develop, compare, or revise their own strategies (PSTs are prompted to justify, explain, or compare mathematical relationships and strategies).	☐	☐	☐	
	The emphasis is on reasoning processes rather than merely obtaining the correct answer.	☐	☐	☐	
2. Computational thinking components	PSTs are encouraged to break the problem into smaller and manageable parts.	☐	☐	☐	
	PSTs are invited to identify, imitate, extend, or modify patterns.	☐	☐	☐	
	The activity provides opportunities for abstraction, and generalisation.	☐	☐	☐	
	The activity enables PSTs to detect, explain, and correct errors.	☐	☐	☐	
	The activity supports PSTs’ understanding and use of algorithmic elements, such as sequencing, loops, and conditionals.				
3. Task progression and levels of challenge	The activity includes simple, extended, and advanced versions of the same core task.	☐	☐	☐	

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4. PSTs choice and flexibility	The activity gives PSTs meaningful choices between alternative solution pathways.	☐	☐	☐	
	PSTs can decide whether to use a digital tool, a non-digital strategy, or a combination of both.	☐	☐	☐	
5. Scaffolding and guidance	Guidance is adjusted to PSTs’ mathematical and technical prior knowledge.	☐	☐	☐	
	The activity provides prompts that support reasoning without giving away the solution.				
	Mathematical, computational, and technical terms are used clearly and consistently throughout the activity.				
6. Reflection and Making Thinking Visible	The activity promotes reflective questioning as a classroom norm, including questions such as “Why does this work?”, “How do I know this is correct?”, “Is this the most efficient approach?”, and “Can you find another way?”	☐	☐	☐	
	PSTs are asked to predict, test, explain, revise and compare their ideas.				
	PSTs document their strategies, decisions, assumptions, representations, errors, and revisions.				
	The activity makes errors, revisions, and debugging processes visible for reflection and discussion.				
	The activity supports classroom discussion of alternative approaches and learning paths.				
	PSTs use documented evidence of their thinking to reflect on and assess their problem-solving processes.				
	Errors, revisions, and debugging processes are made visible for reflection and discussion.				